

PRODUCT DATA CLEANING AND PREPARATION USING MICROSOFT EXCEL

Introduction

This project focuses on performing data cleaning and preparation on a product dataset using Microsoft Excel. The primary objective is to improve data quality and prepare the dataset for meaningful analysis by handling missing values, correcting inconsistencies, removing duplicate records, restructuring columns, and applying formatting techniques.

In real-world business environments, datasets often contain incomplete, duplicated, or inconsistent information that can affect decision-making and analytical accuracy. This project demonstrates how Microsoft Excel can be effectively used to clean, standardize, and organize product-related data for improved usability and interpretation.

Objectives of the Project

The main objectives of this project are:

- To identify and handle missing values in the dataset
- To standardize inconsistent text formats and correct typographical errors
- To identify and remove duplicate records
- To restructure dataset columns through splitting and merging operations
- To apply number and date formatting for better readability
- To implement conditional formatting techniques for improved data visualization

Dataset Description

The dataset used in this project contains the following attributes:

- Product ID
- Product Name
- Brand Name
- Quantity
- Category
- Price

The dataset was analyzed and cleaned to ensure consistency, accuracy, and proper formatting for business analysis purposes.

Data Cleaning and Preparation Process

Handling Missing Values

Missing values were identified and treated appropriately to maintain dataset consistency.

Tasks Performed:

- Checked missing values in the Price column
- Replaced missing price values using average-based methods
- Identified missing entries in the Category column
- Applied suitable methods to standardize missing categories

Formula Used:

"=AVERAGE(range)"

Possible Formula Errors:

- "#DIV/0!" – Occurs when no numerical values are available for averaging
- "#VALUE!" – Occurs if invalid text values are included in calculations

Correcting Inconsistent Data

Text inconsistencies and typographical errors were identified and corrected to maintain data consistency.

Tasks Performed:

- Standardized inconsistent text formats in the Product Name column
- Corrected typographical errors in the Category column
- Used Find and Replace functionality for standardization

Excel Features Used:

- Find and Replace ("Ctrl + H")
- Text formatting techniques

Removing Duplicate Records

Duplicate records were identified and removed to improve data accuracy and avoid redundancy.

Tasks Performed:

- Checked for duplicate entries within the dataset
- Removed duplicate records while preserving original information

Excel Feature Used:

"Remove Duplicates"

Splitting and Merging Data

Column restructuring techniques were applied for better organization of the dataset.

Tasks Performed:

- Split the Product ID column into Manufacturing Date and Country Code
- Removed unnecessary characters where required
- Merged Brand Name and Product Name into a new column named Product Brand

Formula Used:

"=A2&" "&B2"

Number and Date Formatting

Formatting techniques were applied to improve readability and presentation quality.

Tasks Performed:

- Converted the Price column into Currency Format
- Formatted the Manufacturing Date column into DD-MM-YYYY format

Conditional Formatting

Conditional formatting techniques were applied to visually highlight important information.

Tasks Performed:

- Applied Data Bars or Color Scales in the Price column
- Highlighted records where the Category is Electronics using a custom rule

Tools and Features Used

The following Microsoft Excel tools and features were utilized during this project:

- Data Cleaning Techniques
- Find and Replace
- Remove Duplicates
- Text to Columns

- Merge Columns
- Currency Formatting
- Date Formatting
- Conditional Formatting

Product ID	Manufacturing Date	Country Code	Product Name	Brand Name	Brand product	Price (\$)	Quantity	Category
28-JAN-US	28-JAN-2025	US	Laptop	Dell	Dell Laptop	\$ 1,000.00	30	Electronics
15-FEB-US	15-FEB-2025	US	Sneakers	Nike	Nike Sneakers	\$ 80.00	15	Fashion
03-MAR-US	03-MAR-2025	US	Coffee Maker	Keurig	Keurig Coffee Maker	\$ 130.00	40	Kitchen
11-APR-US	11-APR-2025	US	smartphone	Samsung	Samsung Smartphone	\$ 900.00	25	Electronics
22-MAY-US	22-MAY-2025	US	Backpack	North Face	North Face Backpack	\$ 70.00	20	Fashion
07-JUN-UK	07-JUN-2025	UK	Headphones	Sony	Sony Headphones	\$ 250.00	45	Electronics
19-JUL-UK	19-JUL-2025	UK	T-shirt	Adidas	Adidas T-Shirt	\$ 30.00	5	Fashion
23-AUG-UK	23-AUG-2025	UK	Blender	Ninja	Ninja Blender	\$ 90.00	35	Kitchen
05-SEP-UK	05-SEP-2025	UK	Tablet	Apple	Apple Tablet	\$ 500.00	50	Electronics
14-OCT-UK	14-OCT-2025	UK	Hiking Boots	Timberland	Timberland Hiking Boots	\$ 130.00	10	Outdoor
17-JUN-IN	17-JUN-2025	IN	Laptop	HP	Hp Laptop	\$ 950.00	25	Electronics
25-NOV-AU	25-NOV-2025	AU	Sneakers	Adidas	Adidas Sneakers	\$ 90.00	40	Fashion
08-DEC-DE	08-DEC-2025	DE	Coffee Maker	Nespresso	Nespresso Coffee Maker	\$ 120.00	35	Electronics
18-FEB-CA	18-FEB-2025	CA	Smartwatch	Fitbit	Fitbit Smartwatch	\$ 150.00	15	Electronics
16-APR-ES	16-APR-2025	ES	Headphones	Bose	Bose Headphones	\$ 250.00	20	Electronics
21-AUG-CA	21-AUG-2025	CA	Laptop Bag	Samsonite	Samsonite Laptop Bag	\$ 50.00	35	Accessories
20-AUG-CN	20-AUG-2025	CN	Smartwatch	Huawei	Huawei Smartwatch	\$ 160.00	15	Electronics
27-JAN-IT	27-JAN-2025	IT	Laptop	Asus	Asus Laptop	\$ 980.00	10	Electronics
01-MAR-UK	01-MAR-2025	UK	Sunglasses	Oakley	Oakley Sunglasses	\$ 150.00	15	Fashion
14-AUG-US	14-AUG-2025	US	Camping Tent	Coleman	Coleman Camping Tent	\$ 0.00	10	Outdoor
14-MAY-RU	14-MAY-2025	RU	Camera	Nikon	Nikon Camera	\$ 700.00	50	Electronics
09-JAN-CA	09-JAN-2025	CA	Microwave	Panasonic	Panasonic Microwave	\$ 80.00	20	Kitchen
19-JUL-BR	19-JUL-2025	BR	Fitness Tracker	Xiaomi	Xiaomi Fitness Tracker	\$ 150.00	30	Fashion
29-SEP-CA	29-SEP-2025	CA	Smartphone	Google	Google Smartphone	\$ 800.00	45	Electronics
03-JUN-CA	03-JUN-2025	CA	Sunglasses	Ray-Ban	Ray-Ban Sunglasses	\$ 150.00	25	Fashion
11-JUL-CA	11-JUL-2025	CA	Blender	Vitamix	Vitamix Blender	\$ 400.00	40	Kitchen
07-MAR-CA	07-MAR-2025	CA	Dress	Zara	Zara Dress	\$ 60.00	30	Fashion
13-APR-CA	13-APR-2025	CA	Toaster	Hamilton	Hamilton Toaster	\$ 40.00	10	Kitchen
24-MAY-CA	24-MAY-2025	CA	Fitness Tracker	Garmin	Garmin Fitness Tracker	\$ 130.00	5	Electronics
02-DEC-CA	02-DEC-2025	CA	Jeans	Levi's	Levi's Jeans	\$ 50.00	50	Fashion
09-JUL-FR	09-JUL-2025	FR	Watch	Casio	Casio Watch	\$ 100.00	20	Accessories



Conclusion

This project successfully demonstrated the process of cleaning and preparing product-related data using Microsoft Excel. By handling missing values, correcting inconsistencies, removing duplicates, restructuring columns, and applying formatting techniques, the dataset was transformed into a more organized and analysis-ready format. The project strengthened practical knowledge of Excel-based data preparation techniques used in real-world business environments.

Acknowledgment

I would like to express my sincere gratitude to the resources and tools that supported the successful completion of this project.

Special thanks to Microsoft Excel for providing powerful spreadsheet functionalities for data cleaning, transformation, and analysis. I also acknowledge GitHub for project hosting and version control support. Additionally, educational learning resources played an important role in improving my understanding of Excel formulas, data cleaning techniques, and error handling methods such as "#VALUE!" and "#DIV/0!".

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